SUBRAMANYASAI GUNDAMRAJU

Hyderabad, Telangana | 7674038311 | sai.gundamraj14@gmail.com

PROFESSIONAL SUMMARY

AI/ML Engineer experienced in building **end-to-end NLP and deep learning systems** and **production forecasting platforms**, from **pretraining and fine-tuning** language models to designing **multilingual RAG pipelines**, **LLM-driven what-if analysis engines** and CNN-based diagnostic tools.

TECHNICAL SKILLS

Languages: Python, Java, SQL

ML/DL Frameworks: Scikit-learn, Hugging Face Transformers, PyTorch, LightGBM, XGBoost, CatBoost, statsmodels, Prophet, SHAP, Optuna

AI/NLP: LLM Fine-Tuning (LoRA, QLoRA), LLM Pretraining, RAG, Hybrid Search (Dense + BM25), Cross-Encoder Reranking, LLM Agents, Prompt Engineering, Instruction Tuning, OCR (Tesseract), Multilingual NLP (Hindi, Gujarati)

Tools & Databases: QdrantDB, ChromaDB, FastAPI, Flask, Dash, Docker, Apache Airflow, MLflow, Pydantic, Groq, Pandas, NumPy, Matplotlib, Seaborn

Concepts: Time-Series Forecasting, What-If & Scenario Analysis, Causal Testing (Granger, VAR), Feature Engineering, Model Evaluation, Data Structures & Algorithms, DBMS

PROFESSIONAL EXPERIENCE

AI/ML Engineer | ***Vivid Minds Technologies Ltd (Intern → Full-time)*** *Apr 2025 – Present*

Auto-Forecast: Autonomous Forecasting & What-If Scenario Platform | *Python, FastAPI, Dash, LightGBM, statsmodels, PyTorch, SHAP, Pydantic, Docker*

- Architected a **no-code forecasting platform**: give it a CSV/Excel file or database table and it runs a **9-stage pipeline** (ingest, EDA, cleaning, validation, model selection, features, training, report) with **one user confirmation**. Runs are **resumable**, jobs are **cancellable subprocesses**, and it handles **multi-million-row ingest**, **1,000+ series panels** (tested on 1,115 stores), intermittent and multi-seasonal data.
- Built a **14-model catalog** (AutoETS/ARIMA/Theta, MSTL, Prophet, Croston/TSB, LightGBM, XGBoost, CatBoost, NHITS, PatchTST, Chronos) chosen by a **rule-engine ladder** on seasonality, intermittency and structural breaks, and ranked by **walk-forward backtests (MASE)** with prediction intervals and a **convex-weight ensemble**.
- Designed **LLM agents** (cleaning, forecast intent, model ranking, report) on open-weight models (Groq/Ollama) that see **computed statistics only, never raw data**. Each has a **Pydantic schema, sanitizer and rule-based fallback**, so it runs with no LLM, and a **validation gate** blocks data loss and variance collapse.
- Built a **What-If Scenario Copilot** for plain-English questions like **"what if Store 5 raises its price?"** or **"what if we lose half the societies?"**. The LLM only **classifies and narrates**; all numbers come from deterministic engines, and it **declines rather than fabricates**:
  - **Driver what-if:** a 4-tier router (A–D) perturbs lead-lag features on the winning ML model, uses a residual shadow model for classical winners, else a correlation-elasticity estimate.
  - **Goal-seek:** solves for the driver change needed to hit a target. **Attribution:** SHAP + Granger causality + fold-stability. **Driver-vs-driver:** lead-lag, Granger, **VAR**.
  - **Structural what-ifs:** add/remove entities (stores, societies, plants) via a **contribution cube**, as smallest/random/largest ranges, with **historical count-elasticity** (Newey-West CIs, power gate).
  - **Per-series (panel) what-if:** extends all of the above to store/society-level questions.
- Added a **governed external-data layer** (Open-Meteo weather, holidays, geocoding): provider registry, **licence gating**, user-confirmed fetches, audit log and **leakage-free exogenous features**.
- Production-ready: **FastAPI + Dash** UI with live per-model progress, **Docker**, accuracy profiles (fast/balanced/max), an evidence-grounded report and **700+ tests**.

Document Exact RAG: Multilingual Grounded Document Q&A | *Python, Qdrant, BM25, bge-m3, Jina Reranker, Tesseract OCR, Groq (Qwen), Flask*

- Built a **hallucination-resistant hybrid RAG** for English, Hindi and Gujarati: **bge-m3 (Qdrant) + BM25**, merged by **Reciprocal Rank Fusion**, reranked by a **cross-encoder**, and answered by Qwen with **page/paragraph citations**. It gives a fixed "not available" reply when the answer is not in the documents.
- **Multilingual ingestion:** PyMuPDF + **Tesseract OCR** (eng+hin+guj) with **Gujarati digit repair**, **token-aware chunking**, page-safe citations, **table rows as "Header: value" chunks** and **hash-based incremental re-indexing**.
- Deterministic **answer guard:** removes spelled-out numbers, **restores the source's numeral script** and **checks every figure against the retrieved text**; Gujarati/Devanagari digits are normalised for keyword search.

- **Evaluated:** benchmarked **5 embedding models** on a real 283-page tender (bge-m3 **0.857 R@1** vs 0.214 for the previous default); **42-question trilingual gold set** gave **95% hit@3, 0 wrong refusals, 0 unverified numbers**. Cut CPU reranking from **4+ min to ~6 s** (~10 s per answer). SQLite audit log, 76 tests.

Additional work at Vivid Minds

- Pretrained a **small language model** and conducted **attention mechanism** comparisons, implementing custom data pre-processing pipelines and training strategies, building foundational expertise applicable to text processing systems.
- Fine-tuned **LLMs** (**Qwen-0.5B, Gemini-3B**) on custom Q/A generation datasets using **LoRA** and **instruction tuning**, achieving contextually accurate question-answer pair generation with reduced **hallucination**, applicable to automated information extraction and content analysis pipelines.
- Built a **multi-target forecasting tool** on **BANAS** dairy cooperative data (**2021–25**), predicting **milk collection volumes, veterinary visit frequency, and cattle feed requirements** using **LightGBM** models with **feature engineering** on historical time-series data.
- Deployed end-to-end **ML pipelines** using **Apache Airflow** for workflow orchestration and **MLflow** for **experiment tracking, model versioning**, and performance monitoring across all three forecast targets.
- Built a **multi-database routing system** using **Gemini-Flash LLM** and **Qdrant vector database**, enabling intelligent **natural language query routing** across heterogeneous network databases with automated **schema alignment** and precise **SQL query generation**, relevant to enterprise network management and e-governance data systems.
- Integrated **LLM reasoning** with metadata parsing and **vector similarity search (RAG)** to achieve high routing accuracy across diverse database schemas, enabling intelligent **document retrieval**.
- Developed an end-to-end **time-series forecasting pipeline** for **stock price prediction** on **AAPL, Microsoft, and Google** datasets using a hybrid architecture combining two forecasting models, with comprehensive **model evaluation** and performance benchmarking.

Deep Learning Research Intern | ***Centre for AI and ML, Osmania University*** *May 2024 – July 2024*

- Designed and implemented a **UNet generator** with **PatchGAN discriminator** architecture for **face sketch-to-photo synthesis**, enhanced with **attention** and **residual blocks** for improved feature extraction, applicable to **facial recognition** and **video surveillance analytics**.

PROJECTS

MedDetect AI | *Python*, *TensorFlow*, *Keras*, *VGG16*, *CNN*

- Built a **deep learning-based diagnostic system** for **medical imaging** (**X-ray** and **MRI**), classifying image types (chest, brain, knee, hand) and identifying conditions including **pulmonary diseases, brain tumors, osteoporosis,** and **fractures**, demonstrating CNN-based pattern recognition for health management information systems.

EDUCATION

B.Tech Computer Science and Engineering | ***Osmania University*** *May 2025*

CGPA: 9.18/10 | Percentage: 86.34%

ADDITIONAL

- Content creator and script writer for YouTube channel **"Movies4u,"** contributing film reviews and analytical commentary.